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4. $V_1 = 7.05\ dm^3$ $V_2 = 2.35\ dm^3$

$\eta_1 = 0.312$ mol $\eta_2 = ?$

'P' மற்றும் 'T' மாறிலிகள்

$$\therefore \frac{V_1}{n_1} = \frac{V_2}{n_2}$$

$$\Rightarrow n_2 = \left(\frac{n_1}{V_1}\right) \times V_2$$

$$n_2 = \frac{0.312\ mol}{7.05\ \cancel{dm^3}} \times 2.35\ \cancel{dm^3}$$

$$n_2 = 0.104\ mol$$

வெளியிடப்படும் மோல்களின் எண்ணிக்கை

= 0.312-0.104 = 0.208 மோல்கள்

5) $T_1 = 8\ ^0C = 8 + 273 = 281\ K$

$P_1 = 6.4$ atm $V_1 = 2.1$ mol

$T_2 = 25^0\ C = 25 + 273 = 298\ K$

$P_2 = 1$ atm $V_2 = ?$

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$\Rightarrow V_2 = \left(\frac{P_1V_1}{T_1}\right) \times \frac{T_2}{P_2}$$

$$= \frac{6.4\ \cancel{atm} \times 2.1\ mol}{281\ \cancel{K}} \times \frac{298\ \cancel{K}}{1\ \cancel{atm}}$$

$$V_2 = 14.25\ ml$$

6(அ)

$Vo_2 = 12\ dm^3$ $T = 298$, $P = 1$ atm } Constant

$V_{He} = 46\ dm^3$

$V_{total} = 5\ dm^3$

$$P_{O_2} = x_{O_2} \times P_{total}$$

$$x_{O_2} = \frac{n_{O_2}}{n_{O_2} + n_{He}}$$

$$= \frac{0.54}{0.54 + 2.05}$$

$$= \frac{0.54}{2.59} = 0.21$$

$$n_{O_2} = \frac{1\ mol}{22.4\ \cancel{L}} \times 12\ \cancel{L}$$

$$n_{O_2} = 0.54\ mol$$

$$n_{H_e} = \frac{1\ mol}{22.4\ \cancel{L}} \times 46\ \cancel{L}$$

$$n_{H_e} = 2.05\ mol$$

$$P_{total} \times V_{total} = 1\ atm \times 22.4\ l$$

$$\therefore P_{total} = \frac{1\ atm \times 22.4\ \cancel{l}}{5\ \cancel{l}}$$

$$P_{total} = 4.48\ atm$$

$$\therefore P_{O_2} = 0.21 \times 4.48\ atm$$

$$= 0.94\ atm$$

$$P_{He} = x_{He} \times P_{total}$$

$$x_{He} = \frac{n_{He}}{n_{O_2} + n_{He}}$$

$$= \frac{2.05}{0.54 + 2.05}$$

$$x_{He} = \frac{2.05}{2.59} = 0.79$$

$$\therefore P_{He} = 0.79 \times 4.48\ atm$$

$$P_{He} = 3.54\ atm$$

6 (b) $2KClO_{(S)} \longrightarrow 2KCl_{(S)} + 3O_{2(g)}$

$P_{total} = 772$ mm Hg

$P_{H_2O} = 26.7$ mm Hg

$$P_{total} = P_{O_2} + P_{H_2O}$$

$$\therefore P_{O_2} = P_{total} - P_{H_2O}$$

$P_1 = 26.7$ mm Hg $T_2 = 295\ K$